

Exam Question: User Analysis and Persona Development

Scenario: Smart Campus Mobile Application

A large university plans to launch a **Smart Campus Mobile Application** to improve student life and campus operations. The app will integrate multiple services such as:

- Course registration and timetable management
- Campus navigation with real-time directions
- Cafeteria ordering and digital payments
- Library booking and digital resources
- Event discovery and registration
- Shuttle tracking
- Study group formation and messaging

The university has a **diverse population**, including undergraduate students, graduate students, part-time learners, international students, faculty members, and working professionals enrolled in evening programs.

Students access campus services in different contexts—for example, rushing between classes, studying late at night, attending online lectures, or visiting campus occasionally.

The university wants the app to be **user-centered**, ensuring it supports different behaviors, motivations, and usage patterns rather than relying only on demographics.

Part A – User Analysis Using the Matrix Technique

You are hired as a UX researcher to analyze potential users of the Smart Campus App.

Task Requirements:

1. **Create THREE 3×3 matrices** using different behavioral axes.

You may choose axes such as (but not limited to):

- Frequent ↔ Occasional users
- Explorative ↔ Goal-oriented
- Social ↔ Individual
- Visitor ↔ Local
- Curious ↔ Engaged

2. Your matrices must answer the following perspectives:

✓ **Matrix 1:** What can users do? (*Features they would prioritize*)

✓ **Matrix 2:** How do users feel? (*Emotions, motivations, expectations*)

✓ **Matrix 3:** What problems do they face? (*Pain points and frustrations*)

3. For **each cell**, include:

- Likely user behavior
- Key needs or goals
- Possible feature expectations

4. Clearly label insights as:

- **Known** (based on reasonable university context)
- **Assumed** (logical but not verified)
- **Unknown** (requires research)

5. After creating the matrices, write a short analysis identifying:

- Emerging user patterns
- High-priority research areas
- Features important to multiple user groups
- Any risky assumptions

Expected Length: Diagrams + ~400–600 words analysis.

Part B – Persona Development

Using insights from your matrices, create **SIX detailed user personas** for the Smart Campus App.

Requirements:

Each persona must include the following template components:

- ✓ **Name & Photo** (placeholder description is acceptable)
- ✓ **Demographics** (age, role, relevant background)
- ✓ **Goals** (what they want to achieve using the app)
- ✓ **Frustrations** (current problems with campus systems)
- ✓ **Behaviors** (how they currently interact with services)

- ✔ Technology Comfort Level
- ✔ Quote (captures their mindset)

Important Conditions:

- Personas must represent **distinct user groups** (avoid repetition).
- Include a mix of users such as:
 - Highly engaged vs low-engagement users
 - Tech-savvy vs low-tech users
 - On-campus vs remote users
 - Social vs independent users

After creating the personas, briefly explain (150–250 words):

- Which persona should be the **primary target user** and why
- Which personas are **secondary users**
- One persona you would **not prioritize initially**

ANSWER

- ✔ Part A – User Analysis Using the Matrix Technique
- ✔ Matrix 1: What Can Users Do? (Features They Prioritize)

Axes Chosen:

- Horizontal: **Frequent ↔ Occasional**
- Vertical: **Explorative ↔ Balanced ↔ Goal-Oriented**

	Frequent Users	Moderate Users	Occasional Users
Explorative	Try new campus features, join events, explore clubs. <i>(Assumed)</i>	Browse events sometimes. <i>(Assumed)</i>	Check campus map when visiting. <i>(Known)</i>
Balanced	Use timetable + cafeteria regularly. <i>(Known)</i>	Use library booking before exams. <i>(Known)</i>	Open app mainly for announcements. <i>(Assumed)</i>
Goal-Oriented	Quickly check schedule, shuttle, deadlines. <i>(Known)</i>	Complete specific tasks like fee payment. <i>(Known)</i>	Register courses once per semester. <i>(Known)</i>

- ✔ Matrix 2: How Do Users Feel? (Emotions & Motivations)

Axes Chosen:

- Horizontal: **Social ↔ Neutral ↔ Individual**
- Vertical: **Curious ↔ Comfortable ↔ Highly Engaged**

	Social	Neutral	Individual
Curious	Excited to join communities. <i>(Assumed)</i>	Interested but cautious. <i>(Assumed)</i>	Prefer exploring alone. <i>(Assumed)</i>
Comfortable	Feel connected via study groups. <i>(Known)</i>	Expect smooth experience. <i>(Known)</i>	Want minimal interaction. <i>(Known)</i>
Highly Engaged	Motivated by collaboration tools. <i>(Known)</i>	Focused on productivity. <i>(Known)</i>	Value efficiency and control. <i>(Known)</i>

- ✔ Matrix 3: What Problems Do They Face? (Pain Points)

Axes Chosen:

- Horizontal: **Visitor ↔ Semi-local ↔ Local**
- Vertical: **Low Tech ↔ Moderate Tech ↔ High Tech**

	Visitor	Semi-local	Local
Low Tech	Struggle with navigation and login. <i>(Known)</i>	Confused by too many features. <i>(Assumed)</i>	Avoid app due to complexity. <i>(Unknown)</i>
Moderate Tech	Need simple UI guidance. <i>(Known)</i>	Want faster performance. <i>(Known)</i>	Annoyed by bugs. <i>(Known)</i>
High Tech	Expect personalization. <i>(Assumed)</i>	Want integrations (calendar, AI). <i>(Assumed)</i>	Frustrated if app lacks advanced tools. <i>(Assumed)</i>

✓ Analysis & Key Insights (Approx. 500 Words)

After reviewing the matrices, several behavioral patterns emerge.

First, **goal-oriented frequent users** appear across multiple matrices. These users rely heavily on the app for daily academic activities such as checking schedules, booking resources, and tracking transportation. Because they interact with the system regularly, performance, reliability, and speed are critical features. Any downtime or slow loading would significantly impact their productivity.

Second, there is a clear distinction between **explorative users** and **task-focused users**. Explorative users are motivated by campus engagement—events, clubs, and networking opportunities—while goal-oriented users prioritize efficiency. This suggests the app should support both discovery (e.g., personalized event recommendations) and quick navigation (e.g., dashboard shortcuts).

Another important pattern is related to **technology comfort**. Low-tech users struggle with complex interfaces, indicating a need for intuitive design, onboarding tutorials, and accessibility features. Meanwhile, high-tech users expect advanced capabilities such as integrations with calendars, AI reminders, and customization. Designing a flexible interface that supports both beginner and advanced modes could help satisfy these contrasting needs.

The matrices also reveal several **research priorities**. For example, it is unknown whether local low-tech users avoid campus apps entirely or adapt over time. Understanding this could influence whether the design should emphasize training resources or simplify functionality. Additionally, assumptions about personalization features should be validated through user interviews.

Some features appear valuable across multiple user groups, making them high priority for the product roadmap:

- Timetable and academic management
- Campus navigation
- Notifications and announcements
- Library booking
- Shuttle tracking

These features support both frequent and occasional users, ensuring broad adoption.

Risky assumptions include believing all high-tech users want advanced features or assuming occasional users only need minimal functionality. Research may reveal hidden expectations—for example, occasional users might value emergency alerts or exam reminders.

Overall, the analysis suggests the Smart Campus App should prioritize **usability, speed, and essential academic tools first**, followed by engagement features. Designing for flexibility rather than a single “average user” will ensure the app supports the diverse campus population effectively.

✓ Part B – Six User Personas

Persona 1: Academic Achiever Ayesha

Photo: Organized student carrying a laptop

Demographics:

- Age: 20
- Undergraduate student
- Lives in university hostel

Goals:

- Track classes and deadlines
- Book library rooms
- Stay academically organized

Frustrations:

- Slow apps waste time
- Too many notifications

Behaviors:

- Checks timetable daily
- Plans her week in advance

Tech Comfort: High

Quote:

"I don't have time for apps that slow me down."

Persona 2: Social Networker Saad

Photo: Friendly student attending events

Demographics:

- Age: 21
- Business major
- Very active on campus

Goals:

- Discover events
- Join clubs
- Meet new people

Frustrations:

- Missing event updates
- Poor messaging tools

Behaviors:

- Opens apps multiple times daily
- Shares events with friends

Tech Comfort: High

Quote:

"University is more than classes—it's about connections."

Persona 3: Budget-Conscious Bilal

Photo: Student comparing prices on phone

Demographics:

- Age: 22
- Part-time worker + student

Goals:

- Order affordable food
- Track expenses
- Find discounts

Frustrations:

- Hidden costs
- Complicated payment systems

Behaviors:

- Always compares prices
- Uses digital wallets

Tech Comfort: Moderate

Quote:

"Every rupee matters."

Persona 4: Busy Professional Priya

Photo: Working professional with backpack

Demographics:

- Age: 29
- Evening MBA student
- Full-time employee

Goals:

- Quickly access course materials
- Navigate campus efficiently
- Avoid wasting time

Frustrations:

- Long login processes
- Information buried in menus

Behaviors:

- Uses app only when necessary
- Prefers shortcuts

Tech Comfort: High

Quote:

"Give me what I need in seconds."

Persona 5: International Student Ibrahim

Photo: New student holding campus map

Demographics:

- Age: 24
- Exchange student
- Recently arrived

Goals:

- Learn campus layout
- Understand university systems
- Connect with peers

Frustrations:

- Confusing processes
- Lack of guidance

Behaviors:

- Frequently checks maps
- Searches FAQs

Tech Comfort: Moderate

Quote:

"Everything is new—I need guidance."

Persona 6: Minimalist Maryam

Photo: Quiet student studying alone

Demographics:

- Age: 23
- Graduate researcher

Goals:

- Access library resources
- Avoid distractions

Frustrations:

- Social notifications
- Feature overload

Behaviors:

- Uses only essential tools
- Turns off alerts

Tech Comfort: High

Quote:

"Simple is always better."

✓ Persona Prioritization (Approx. 200 Words)

The **primary target persona** should be **Academic Achiever Ayesha** because she represents a large portion of the university population—full-time students who depend heavily on campus systems. Designing for her ensures the app supports core academic workflows, which are critical for adoption.

Secondary personas include Social Networker Saad and International Student Ibrahim. Saad drives engagement and campus participation, helping increase daily app usage, while Ibrahim highlights the need for intuitive navigation and onboarding—features that improve usability for everyone.

Busy Professional Priya is also important but represents a smaller segment. Her needs emphasize efficiency and could guide the development of quick-access dashboards.

The persona that should **not be prioritized initially** is Minimalist Maryam. While her preference for simplicity is valuable, her limited interaction means she is less likely to influence early adoption metrics. However, her needs should still inform clean interface design to prevent feature overload.

By focusing first on academically driven frequent users while ensuring usability for newcomers, the Smart Campus App can achieve both high adoption and long-term satisfaction.